A Compact CNN-LSTM Network Intrusion Detection System with Leave-One-Attack-Out Generalization Analysis on CICIDS2017

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Abstract

Network Intrusion Detection Systems (NIDS) based on deep learning have reported near-ceiling accuracies on the CICIDS2017 benchmark, with published CNN-LSTM and hybrid architectures routinely exceeding 99% under conventional random-split evaluation. We argue, and empirically demonstrate, that this evaluation protocol is a poor proxy for real-world robustness: a model can achieve arbitrarily high accuracy while having learned to recognize only the specific attack categories present at training time, with no guarantee of behavior on a category it has never encountered – precisely the scenario a zero-day attack presents. In this work we first build a compact CNN-LSTM classifier (33,761 parameters, roughly an order of magnitude smaller than typical published architectures) that reaches 98.29% test accuracy and 0.9989 ROC-AUC under standard evaluation, with a train-test accuracy gap of 0.0005 and a 5-fold cross-validation standard deviation of 1.39%, indicating low bias and low variance under this protocol. We then subject the same model to a stricter Leave-One-Attack-Out (LOAO) evaluation, in which an entire attack category is withheld from training and treated as unseen at test time, across four categories (DoS Hulk, PortScan, DDoS, Bot). Accuracy falls to 63–80% depending on category, with a complete recall failure (0.00%) on Bot traffic – a failure mode that, notably, exactly matches an independent finding by Xu and Liu (2025) on a different CNN architecture and category set, suggesting a structural rather than implementation-specific cause. We provide a distribution-shift argument, grounded in empirical risk minimization theory, for why this collapse is expected rather than anomalous. We then test whether two classifier-independent signals – embedding-space distance to class centroids and benign-only autoencoder reconstruction error – can recover any of this lost performance, via a formal four-condition ablation. We find negligible improvement (under 0.3 percentage points in every configuration), a negative result we use to localize the true bottleneck to the learned representation itself rather than to the decision rule applied on top of it. On standard evaluation our model is competitive with, though does not exceed, the strongest published figures (99.49–99.97%), while using substantially fewer parameters; under LOAO evaluation, to our knowledge no compared work reports this protocol at all, making direct superiority claims inapplicable rather than favorable or unfavorable. We conclude with a concrete, mathematically motivated direction for closing this gap: replacing centroid comparison with instance-level retrieval and training the embedding space itself under a metric-learning objective rather than plain cross-entropy classification.

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1 Introduction

1.1 Motivation

Network intrusion detection sits at the front line of modern cybersecurity infrastructure, tasked with distinguishing malicious traffic from the overwhelming volume of benign network activity in real time. The rise of deep learning has reshaped this field: hybrid architectures combining convolutional layers (for local statistical feature extraction across flow-level attributes) with recurrent layers such as LSTM (for compositional feature aggregation across a flow's temporal structure) have become a dominant paradigm for NIDS research on the CICIDS2017 benchmark, with published results frequently exceeding 99% accuracy [2, 3]. On the surface, this suggests the problem of network intrusion detection is close to solved.

We argue this conclusion is premature, and that the near-ceiling accuracy figures dominating the literature reflect a property of the evaluation protocol rather than of the underlying detection capability. Under a standard random train/test split, every attack category present in the test set was also present, in different specific instances, during training. A model achieving 99% accuracy under this protocol has demonstrated that it can generalize *within* the distribution of attack behaviors it was shown – a meaningfully narrower claim than generalizing to *novel* attack behavior, which is the actual threat model a deployed NIDS must contend with. A zero-day attack is, by definition, a category absent from any training corpus available at the time the model was built. No amount of accuracy on known categories logically implies anything about behavior on an unknown one; this is the central methodological gap this work is built to probe.

1.2 Problem Formalization

Let $f_\theta : \mathcal{X} \to [0, 1]$ denote the trained classifier, parameterized by θ, fit via empirical risk minimization over a training set $\{(x_i, y_i)\}_{i=1}^n$ drawn i.i.d. from a joint distribution $P_{\text{train}}(x, y)$:

$$\theta^* = \arg\min_\theta \; \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}\big(f_\theta(x_i), y_i\big) \; + \; \lambda_1 \|\theta\|_1 \; + \; \lambda_2 \|\theta\|_2^2 \tag{1}$$

where $\mathcal{L}$ is binary cross-entropy and the ℓ_1/ℓ_2 penalty terms constrain the effective hypothesis space, directly trading a small increase in bias for a reduction in variance – the mechanism underlying the low train–test gap reported in Section 5. Standard generalization guarantees for this procedure (e.g., PAC-style bounds) hold under the assumption that test examples are drawn from the same distribution P_{train} as the training data. Under conventional random-split evaluation, this assumption is satisfied by construction: every category contributing to the test distribution also contributed, independently, to the training distribution.

The Leave-One-Attack-Out (LOAO) protocol we adopt deliberately violates this assumption. Let $\mathcal{C} = \{c_1, \ldots, c_k\}$ denote the set of attack categories, and let $P_{c_j}(x \mid y = 1)$ denote the conditional distribution of flow features for category c_j. Under LOAO, the model is trained on $P_{\text{train}}(x, y \mid \text{category} \neq c_j)$ for some withheld c_j, and evaluated on test examples drawn from $P_{c_j}(x \mid y = 1)$ – a distribution whose support the model has never observed under the positive class during training. No portion of the standard ERM generalization argument applies to this regime; the model's behavior on P_{c_j} is, formally, extrapolation outside its training support, with no theoretical guarantee of correctness. We treat the resulting accuracy collapse (Section 6.4) not as an anomaly requiring explanation away, but as the *expected* consequence of this distribution mismatch – and we use this framing to motivate why our subsequent augmentation attempt (Section 6.2) targets the representation itself, since distance-based or reconstruction-based scoring rules computed within

an embedding space shaped entirely by P_{train} inherit the same support limitation. An embedding function $g_\theta : \mathcal{X} \to \mathbb{R}^d$ (here, the model's penultimate 32-dimensional layer) is itself a learned function of θ^*, and θ^* was optimized with no exposure to P_{c_j}; there is no reason, a priori, to expect distances computed in $g_\theta(\mathcal{X})$ to separate P_{c_j} from P_{benign} any more reliably than f_θ itself does.

1.3 Contributions

This work makes the following contributions:

1. A compact, parameter-efficient CNN-LSTM classifier (33,761 parameters) evaluated under the standard CICIDS2017 protocol, with full bias-variance diagnostics (5-fold cross-validation, train/validation/test gap analysis) and a comparison against published state-of-the-art results.

2. A Leave-One-Attack-Out evaluation across four withheld categories, quantifying the generalization gap between standard and zero-day-style evaluation that is absent from the compared literature [2, 3].

3. A formal, component-wise ablation testing whether embedding-distance and reconstruction-error signals can recover LOAO performance, reported as a negative result with an explicit distribution-shift explanation for why this outcome is theoretically expected rather than a failure of implementation.

4. A cross-study convergence check against an independently reported CNN zero-recall failure on Bot traffic [4], strengthening the claim that this specific failure mode is structural to the dataset and supervised paradigm rather than an artifact of either individual implementation.

1.4 Summary of Findings Relative to Prior Work

To state plainly where this work stands relative to the literature it compares against: under standard evaluation, our model does not exceed the highest published accuracy (99.97%, [3]) or the strongest published F1 (0.9956, [2]); it performs comparably, within the same 97–99.9% tier reported across CICIDS2017 studies, while using roughly an order of magnitude fewer parameters than either compared architecture, neither of which reports its parameter count. Under LOAO evaluation, direct comparison is not possible in the strict sense, since neither [2] nor [3] report results under this protocol at all – our contribution here is introducing this harder evaluation to a setting where it has not previously been applied, rather than outperforming a directly comparable number. Where a LOAO-style comparison *is* available, against Xu and Liu's unknown-attack evaluation [4], our aggregate LOAO accuracy (70.66%) is broadly comparable in magnitude to their reported figure (58.82%), and our independent replication of their exact 0.0000 Bot-recall failure is, we argue, a more valuable finding than a favorable accuracy comparison would be, since it is evidence of a shared, structural limitation rather than an implementation-specific weakness in either study. We emphasize that we do not claim to outperform any individually comparable published figure, category-for-category; where a direct category-level comparison exists (Section 2), our result is either behind it or matches a known failure mode. Our augmentation ablation, finally, is an explicit negative result: neither centroid-distance nor reconstruction-error signals improve on the raw classifier's baseline LOAO performance, which we argue is itself an informative finding about where the true bottleneck in zero-day NIDS generalization lies.

1.5 Paper Organization

Section 2 reviews related work on CICIDS2017 classification and zero-day/unseen-attack detection. Section 3 describes the dataset and preprocessing pipeline. Section 4 details the proposed architecture. Section 5 reports standard-evaluation results and the comparative analysis against published state-of-the-art methods. Section 6 presents the core contribution of this work: the LOAO evaluation, the augmentation ablation, and a mathematically grounded discussion of the negative result. Section 7 discusses deployment implications, and Section 9 concludes with limitations and concrete directions for future work.

2 Related Work

2.1 Supervised CNN-LSTM Architectures on CICIDS2017

Several published CNN-LSTM and hybrid architectures report near-ceiling accuracy on CICIDS2017 under standard evaluation. Afraji et al. [2] propose a parallel-fusion architecture that extracts spatial features via CNN, short-term temporal features via GRU, and long-term temporal dependencies via LSTM in parallel branches before fusion, achieving 99.49% accuracy, 99.25% precision, and 99.88% recall on CICIDS2017 – the strongest published F1-score (0.9956) among the works surveyed for this comparison. Kumar and Sharma [3] instead emphasize feature engineering: an intelligent hybrid feature selection stage reduces the input space to 27 discriminative features before classification, reaching 99.967% accuracy, the highest published figure surveyed here. Neither work reports performance under a held-out-category (zero-day) evaluation protocol; both are evaluated exclusively under random-split conditions where every attack category represented in the test set is also represented, by other instances, in the training set.

2.2 Autoencoder-Based and Unsupervised Zero-Day Detection

A separate line of work sidesteps the held-out-category problem by never training on attack labels at all. Hindy et al. [1] train an autoencoder exclusively on benign traffic and use reconstruction error, thresholded at several operating points (0.05, 0.10, 0.15), as an anomaly signal to flag zero-day attacks. Their per-category detection accuracy on CICIDS2017 varies substantially by attack type: attacks with large statistical separation from benign traffic (e.g., DoS Hulk, DDoS) are detected with 92–99% accuracy even at conservative thresholds, whereas attacks with subtler, benign-mimicking behavior (e.g., Botnet, PortScan at strict thresholds) are detected far less reliably, in some configurations below 20% before threshold relaxation. This category-dependent difficulty is a recurring theme that our own LOAO results reproduce independently under a supervised rather than unsupervised paradigm, suggesting the difficulty is a property of the attack categories themselves rather than of any single detection methodology.

This general finding – that anomaly-detection paradigms trained without attack labels can sometimes generalize better to unseen threats than purely supervised classifiers, at the cost of elevated false-alarm rates – is consistent with the broader anomaly detection literature [5], and with evaluation methodologies developed for related network security benchmarks such as UNSW-NB15 [6].

2.3 Controlled Known-vs-Unknown Attack Comparisons

Most directly relevant to the present work, Xu and Liu [4] conduct a controlled comparison of supervised (MLP, CNN) and unsupervised (OCSVM, LOF) models on CICIDS2017 under both

a standard evaluation and an explicit unknown-attack test set (DoS slowloris, DoS Slowhttptest, and Bot withheld from training). Their supervised CNN model achieves 96.50% accuracy on the standard test set but collapses to 58.82% accuracy and 19.54% recall on the unknown-attack set. Critically, their reported per-class accuracy for CNN on the withheld Bot category is exactly 0.0000 – an identical total failure mode to what we independently observe for Bot in our own LOAO evaluation (Section 6.4). This convergent finding across two independent studies, on the same dataset, using different CNN architectures and different withheld category sets, is strong evidence that the Bot-detection failure is a structural property of the dataset and supervised learning paradigm, not an artifact of a single implementation. We discuss the theoretical basis for why such convergence is expected, rather than coincidental, in Section 6.5.

2.4 Positioning of This Work

The literature reviewed above splits into two largely non-overlapping camps: works that push standard-evaluation accuracy toward its ceiling without testing generalization to unseen categories [2, 3], and works that study unseen-category generalization but do not combine it with a rigorous component-wise ablation of candidate remedies [1, 4]. This work sits at the intersection: it builds a standard-evaluation-competitive classifier, subjects it to the harder protocol, and – rather than stopping at reporting the collapse – formally tests, with an isolated ablation rather than a single before/after number, whether two plausible remedies recover any of the lost performance.

3 Dataset and Preprocessing

We use the CICIDS2017 dataset, comprising 2,830,743 labeled network flow records across 15 classes (1 benign, 14 attack sub-types), generated by the Canadian Institute for Cybersecurity to reflect realistic enterprise network traffic. The raw class distribution is severely imbalanced: Benign traffic accounts for 85.46% of records (1,977,318 samples), while the rarest class, Heartbleed, contains only 11 samples – an imbalance ratio of approximately 179,756:1. The full label distribution is summarized in Table 1.

Table 1: CICIDS2017 class distribution (top classes by frequency).

Class	Count	% of Total
Benign	1,977,318	85.457
DoS Hulk	172,846	7.470
DDoS	128,014	5.533
DoS GoldenEye	10,286	0.445
FTP-Patator	5,931	0.256
DoS slowloris	5,385	0.233
DoS Slowhttptest	5,228	0.226
SSH-Patator	3,219	0.139
PortScan	1,956	0.085
Bot	1,437	0.062
Heartbleed	11	0.000

3.1 Preprocessing Pipeline

For standard classification, the 14 attack sub-types are collapsed to a single "Attack" label, yielding a binary Benign/Attack task, and the dataset is balanced-sampled for computational tractability. All flow-level features are standardized to zero mean and unit variance using `StandardScaler`, fit exclusively on the training partition to prevent information leakage from validation or test distributions into the scaling statistics:

$$\hat{x}_{i,k} = \frac{x_{i,k} - \mu_k^{\text{train}}}{\sigma_k^{\text{train}}} \tag{2}$$

where μ_k^{train} and σ_k^{train} are the mean and standard deviation of feature k computed only over training samples. Standardized features are reshaped to a (features, 1) sequence format to serve as input to the Conv1D front end. A stratified 70/15/15 train/validation/test split is used for the standard-evaluation experiments, and an independent 5-fold stratified cross-validation is run separately for the bias-variance diagnostic reported in Section 5.

3.2 LOAO-Specific Partitioning

For the Leave-One-Attack-Out experiments (Section 6), a separate partitioning procedure is required. For a given withheld category c_j, all records labeled c_j are removed from the training and validation pools entirely; the training set retains benign traffic and every *other* attack category. The test set is constructed from held-out benign traffic plus every record of the withheld category c_j, so that the test distribution for the positive class is $P_{c_j}(x \mid y = 1)$ exactly, with no leakage of c_j instances into any portion of training. This partitioning is repeated independently for each of the four withheld categories evaluated in this work (DoS Hulk, PortScan, DDoS, Bot), with a fresh model trained from random initialization for each.

4 Proposed Architecture

The classifier is a compact hybrid CNN-LSTM with two convolutional blocks feeding a single LSTM layer, summarized in Table 2.

Table 2: Proposed CNN-LSTM architecture.

Layer
Conv1D(64 filters, kernel=3, ReLU, L1L2 regularized)
BatchNormalization
MaxPooling1D(pool=2)
Conv1D(32 filters, kernel=3, ReLU, L1L2 regularized)
BatchNormalization
MaxPooling1D(pool=2)
LSTM(64 units)
Dropout(0.3)
Dense(32, ReLU)
Dropout(0.3)
Dense(1, sigmoid)

4.1 Layer-Level Formulation

Each one-dimensional convolution computes, for input sequence x, kernel w of size K, and bias b:

$$(x * w)[t] = \sum_{k=0}^{K-1} x[t+k]\, w[k] + b, \qquad \text{followed by ReLU}(z) = \max(0, z). \tag{3}$$

Each max-pooling operation reduces resolution and introduces local translation invariance:

$$y[t] = \max_{0 \le k < P} z[t \cdot S + k], \tag{4}$$

where P is the pooling window and S the stride. The LSTM layer updates a hidden state h_t and cell state c_t via the standard gated recurrence:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{5}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{6}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{7}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{8}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{9}$$
$$h_t = o_t \odot \tanh(c_t) \tag{10}$$

which gives the recurrent layer the capacity to aggregate information across the convolved feature sequence while selectively retaining or discarding state, a property the CNN front end alone lacks. The final hidden state is passed through Dropout, a Dense(32) bottleneck – which also serves as the embedding used for the distance-based augmentation in Section 6.2 – a second Dropout, and a final sigmoid unit for binary classification:

$$\hat{y} = \sigma\big(W_{\text{out}} \cdot \text{Dense}_{32}(h_T) + b_{\text{out}}\big) = \frac{1}{1 + e^{-z}}. \tag{11}$$

4.2 Regularization and Parameter Budget

L1/L2 regularization ($\lambda_1 = 10^{-5}$, $\lambda_2 = 10^{-4}$ in Equation 1) is applied to convolutional kernels to control overfitting without materially constraining model capacity. The model totals **33,761 parameters** (33,569 trainable), an order of magnitude smaller than most published CICIDS2017 architectures, which typically do not report parameter counts at all – a property we revisit in Section 7 as a deployment advantage independent of raw accuracy.

4.3 Training Procedure

Training uses the Adam optimizer with `EarlyStopping` (patience 3, restoring best weights on validation loss) and `ReduceLROnPlateau` to anneal the learning rate on validation-loss plateaus. A custom callback tracks test-set loss at every epoch purely for diagnostic plotting; critically, this value never influences any training decision, which depends only on the validation split, preserving the integrity of the held-out test set as a measure of generalization rather than of implicit tuning.

5 Standard Classification Results

Training converged in 19 epochs (batch size 512, 480,544 training samples). Table 3 summarizes the bias-variance diagnostic.

Table 3: Bias-variance diagnostic.

Metric	Value
5-fold CV validation accuracy	0.9721 ± 0.0139
Train accuracy	0.9834
Validation accuracy	0.9827
Test accuracy	0.9829
Train–test accuracy gap	0.0005

The negligible train–test gap (0.05 percentage points) combined with a low cross-validation standard deviation (1.39%) indicates **low bias and low variance**: the model is neither underfitting nor overfitting under the standard evaluation protocol. This diagnostic matters specifically because it rules out a confound in the LOAO results that follow in Section 6: the collapse reported there cannot be attributed to the base model being poorly fit in general, since under the distribution it was actually trained on, it is well-calibrated by every standard bias-variance measure.

Final test-set performance is summarized in Table 4.

Table 4: Test set performance (binary classification, standard split).

Metric	Value
Accuracy	0.9829
Precision (weighted)	0.9829
Recall (weighted)	0.9829
F1-score (weighted)	0.9829
Specificity	0.9782
ROC-AUC	0.9989

5.1 Comparative Analysis with State-of-the-Art

Table 5 places our model against three published results on CICIDS2017.

Table 5: Comparison against published CICIDS2017 results (standard evaluation only).

Model	Accuracy	F1	Parameters
CNN-LSTM-GRU, parallel fusion [2]	0.9949	0.9956	Not reported
Hybrid DL, 27 selected features [3]	0.9997	–	Not reported
Benign-only Autoencoder [1]	0.75–0.99*	–	Not reported
Proposed CNN-LSTM (this work)	**0.9829**	**0.9829**	**33,761**

*Autoencoder detection accuracy is category-dependent (75–99% range across attack types and thresholds); not a single aggregate figure.

On standard evaluation, our model does not exceed the highest published figure (99.97% [3]), which uses a curated 27-feature subset and a more elaborate hybrid pipeline. Our architecture

instead prioritizes efficiency: it uses the full feature set with roughly an order of magnitude fewer parameters than typical published architectures while remaining within the same accuracy tier (97–99.9%) reported across CICIDS2017 studies. We state this comparison without qualification: on this specific metric, under this specific protocol, this work does not claim or demonstrate superiority over [2] or [3]. The contribution of this work lies instead in Section 6, where none of the compared architectures have published results at all.

6 Innovation: Leave-One-Attack-Out Evaluation with Centroid- and Reconstruction-Based Augmentation

6.1 Motivation

A classifier trained to recognize specific attack categories provides no theoretical guarantee that it will recognize a category it has never seen, as formalized in Section 1.2. To test this empirically, we implement a Leave-One-Attack-Out (LOAO) protocol: for each of four attack categories (DoS Hulk, PortScan, DDoS, Bot), we retrain the CNN-LSTM from scratch with that category entirely excluded from the training pool, then evaluate on a test set containing only the withheld category plus held-out benign traffic. This directly measures generalization to a genuinely unseen threat, rather than to a held-out split of an already-seen distribution.

6.2 Augmentation Method

Rather than relying solely on the retrained classifier's decision boundary, we test whether two additional, classifier-independent signals can recover generalization performance.

Centroid-distance features. The 32-dimensional embedding $g_\theta(x) \in \mathbb{R}^{32}$ from the model's penultimate Dense layer is extracted for every training sample. A benign centroid and one centroid per *known* attack category (i.e., excluding the withheld one) are computed as class-wise embedding means:

$$\mu_{\text{benign}} = \frac{1}{|\mathcal{B}|} \sum_{x_i \in \mathcal{B}} g_\theta(x_i), \qquad \mu_c = \frac{1}{|\mathcal{A}_c|} \sum_{x_i \in \mathcal{A}_c} g_\theta(x_i) \;\; \forall c \in \mathcal{C} \setminus \{c_j\} \tag{12}$$

where $\mathcal{B}$ is the benign training set and $\mathcal{A}_c$ is the training set for known attack category c. For each query sample, three features are derived: Euclidean distance to the benign centroid, Euclidean distance to the nearest known-attack centroid, and the signed gap between the two:

$$d_{\text{benign}}(x) = \|g_\theta(x) - \mu_{\text{benign}}\|_2, \qquad d_{\text{attack}}(x) = \min_{c \in \mathcal{C} \setminus \{c_j\}} \|g_\theta(x) - \mu_c\|_2 \tag{13}$$

$$\Delta(x) = d_{\text{benign}}(x) - d_{\text{attack}}(x) \tag{14}$$

The underlying hypothesis is that a genuinely novel attack should embed far from both known clusters, or at least measurably closer to the attack region than to benign.

Benign-only autoencoder reconstruction error. A small autoencoder (input→32→8→32→input, trained with Adam and mean-squared-error loss) is trained exclusively on benign traffic in the raw scaled feature space. For encoder E and decoder D, the reconstruction error on a query sample is:

$$r(x) = \frac{1}{m} \sum_{k=1}^{m} \big(x_k - D(E(x))_k\big)^2 \tag{15}$$

where m is the feature dimensionality. This signal is computed independently of the CNN-LSTM entirely and is intended as a category-agnostic "does this resemble normal traffic" score.

Meta-classifier. These signals are combined with the base classifier's output probability $p(x) = f_\theta(x)$ via a lightweight logistic regression meta-classifier, evaluated under a four-condition ablation so that each component's individual contribution can be isolated:

$$\text{Baseline:} \quad p(x) \geq 0.5 \tag{16}$$

$$\text{Exp1 (A):} \quad \sigma(w_0\, p(x) + b) \tag{17}$$

$$\text{Exp2 (A+B):} \quad \sigma\big(w_0\, p(x) + w_1\, d_{\text{benign}}(x) + w_2\, d_{\text{attack}}(x) + w_3\, \Delta(x) + b\big) \tag{18}$$

$$\text{Proposed (A+B+C):} \quad \sigma\big(w_0\, p(x) + w_1\, d_{\text{benign}}(x) + w_2\, d_{\text{attack}}(x) + w_3\, \Delta(x) + w_4\, r(x) + b\big) \tag{19}$$

This design isolates each component's contribution, rather than reporting a single before/after comparison that would conflate the two signals' individual effects.

6.3 A Note on Terminology

This design is inspired by the same intuition behind retrieval-augmented approaches – using stored knowledge about known categories to inform a new decision – but it is **not** Retrieval-Augmented Generation (RAG). No generative model is used, and there is no instance-level retrieval step: distances are computed against fixed, precomputed class centroids rather than against individually retrieved training examples. A more faithful adaptation of the retrieval paradigm to this classification setting would replace the centroid comparison with a k-nearest-neighbor search over stored training embeddings, using the label distribution of the retrieved neighbors as the augmenting signal. We identify this as a concrete direction for future work (Section 8) rather than claiming it as already implemented here.

6.4 Results

Table 6 reports baseline (no augmentation) LOAO performance per withheld category, and Table 7 reports the full ablation, averaged across all four categories.

Table 6: Baseline CNN-LSTM performance under LOAO evaluation (category fully withheld from training).

Withheld Category	Accuracy	Recall	F1
DoS Hulk	0.6319	0.6215	0.7665
PortScan	0.7984	0.3712	0.5088
DDoS	0.6443	0.6313	0.7736
Bot	0.7519	**0.0000**	**0.0000**

Two findings stand out. First, accuracy collapses substantially relative to the 98.29% standard-split result whenever an entire category is withheld, confirming empirically what Section 1.2 predicts

Table 7: Ablation study, averaged across all four withheld categories. A = base classifier probability, B = centroid-distance features, C = autoencoder reconstruction error.

Condition	Accuracy	Recall	F1
Baseline (A only, direct)	0.7066	0.4060	0.5122
Exp1: A (recalibrated)	0.7063	0.4068	0.5124
Exp2: A+B	0.7053	0.4063	0.5119
Proposed: A+B+C	0.7053	0.4063	0.5119

theoretically: the standard evaluation in Section 5.1 does not reflect zero-day robustness. Second, and central to the ablation's purpose, the added centroid-distance and autoencoder signals produce **negligible change** relative to the baseline classifier probability alone – in most conditions the difference is within 0.1–0.3 percentage points, and on Bot specifically, recall remains exactly 0% across every configuration tested.

6.5 Discussion: Interpreting the Negative Result

We do not present the ablation result as an improvement, and we consider it important to state this plainly rather than overstate the outcome. The value of this experiment lies in what it rules out: the bottleneck limiting zero-day generalization in this setting is not the *scoring rule* applied after classification, since neither embedding-distance nor reconstruction-error augmentation recovers meaningful performance. This points instead to the *representation itself* – as argued formally in Section 1.2, the 32-dimensional embedding space learned by the CNN-LSTM is shaped entirely by the attack categories present during training, via the same optimization (Equation 1) that produces θ^*. Distances measured within that space, even to a well-chosen set of centroids, do not carry reliable information about categories the space was never shaped to separate, because g_θ was never optimized with any objective that would encourage such separation for an unseen class. A reconstruction-error signal computed independently of the classifier fares no better here, though prior work shows autoencoder-based detection *can* succeed when attacks are statistically well-separated from benign traffic [1] – our LOAO setting evidently does not provide that separation for the categories tested.

The complete recall failure on Bot (0.00% in every condition) is not, in isolation, evidence of a flaw specific to our implementation. Xu and Liu [4] independently report an identical 0.0000 per-class accuracy for a CNN model on withheld Bot traffic in CICIDS2017, using a different architecture and a different held-out category set. The convergence of two independent studies on this exact failure mode is stronger evidence that Bot traffic is intrinsically difficult to generalize to under a supervised, fully-withheld evaluation than any single result could provide alone – and is consistent with the distribution-support argument of Section 1.2: if Bot traffic occupies a region of feature space poorly separated from benign traffic even when *included* in training (a property independent of which model is used), then no amount of representational capacity withheld-category retraining can be expected to fix, since the issue is not a withheld-category artifact but a property of the category's statistical signature relative to benign traffic. Averaged across all four withheld categories, our baseline LOAO accuracy (70.66%) is broadly comparable in magnitude to the unknown-attack accuracy Xu and Liu report for their CNN model (58.82%); we note this only as a coarse, order-of-magnitude comparison, since the two studies withhold different category sets and use different sampling and feature pipelines, not as a controlled head-to-head benchmark. We do not interpret this comparison as evidence of superiority in any individual category; where the two

studies' withheld categories overlap (Bot), the result is a tie in total failure, not an advantage for either study.

7 Real-World Applications

The compact parameter count (33,761) makes this architecture a reasonable candidate for resource-constrained deployment contexts – edge gateways, IoT hubs, or embedded network appliances – where a full-scale ensemble or transformer-based IDS would be impractical. On the standard-evaluation task, the model is well-suited to complement a signature-based system as a lightweight secondary classifier for known attack families, operating at low latency with a small memory footprint.

However, the LOAO results in Section 6.4 carry a direct and important deployment implication: this class of model, on its own, should not be relied upon as a primary defense against attack categories not represented in its training data. In an enterprise or cloud Security Operations Center (SOC) context, this motivates a layered deployment strategy – using the CNN-LSTM classifier for known-attack triage (where it performs well, Section 5.1) while retaining human analyst review or a complementary boundary-based method for traffic flagged as anomalous but not confidently classified, particularly for categories with a documented history of poor supervised-model generalization such as Bot and PortScan traffic. The theoretical framing of Section 1.2 generalizes beyond this specific architecture: *any* classifier trained purely by empirical risk minimization over a fixed category set inherits the same support-limitation argument, suggesting this layered-deployment recommendation applies to the wider class of supervised NIDS models, not solely to the architecture studied here.

8 Limitations and Future Work

- **Augmentation did not close the generalization gap.** The centroid-distance and autoencoder signals tested here provided no meaningful recovery of LOAO performance. A concrete next step, motivated directly by this negative result, is to replace centroid-based comparison with genuine instance-level k-nearest-neighbor retrieval over stored training embeddings, and separately, to incorporate a contrastive or metric-learning training objective – for instance, a triplet or supervised contrastive loss added to Equation 1 – so that the embedding space itself is shaped to separate "normal vs. anomalous" rather than "benign vs. these specific known attacks."

- **Cross-study comparisons are approximate.** The comparison to [4] in Section 6.4 uses a different held-out category set, sampling strategy, and feature pipeline than this work, and should be read as directional evidence of a shared phenomenon rather than a controlled benchmark. A fully controlled replication, using an identical withheld category set and preprocessing pipeline across both architectures, is left to future work.

- **Bot detection remains an open problem.** Given its low absolute sample count (1,437 records) and the convergent zero-recall finding across two independent studies, few-shot calibration using a small number of labeled Bot examples – rather than a fully zero-shot setting – may be a more tractable path than further zero-shot augmentation.

- **Only four categories were evaluated under LOAO.** A full leave-one-out sweep across all 14 attack sub-types, while computationally expensive (requiring one full retraining per category),

would give a more complete picture of which categories are and are not tractable under this protocol, and is a natural extension of this work.

9 Conclusion

This work builds a parameter-efficient CNN-LSTM intrusion detector that is competitive with published CICIDS2017 results under standard evaluation, and then subjects it to a stricter, less commonly reported Leave-One-Attack-Out evaluation that more accurately reflects zero-day deployment conditions. Consistent with independently published findings [4], we observe a substantial and, for one category, total collapse in detection performance under this harder protocol. A rigorous ablation of two candidate augmentation signals shows neither meaningfully recovers this lost performance, a negative result that – grounded in the distribution-shift argument of Section 1.2 – localizes the generalization bottleneck to the learned representation rather than the post-hoc scoring method, and motivates a specific, actionable direction – metric-learning-based representation training and genuine instance-level retrieval – for future work. We make no claim of outperforming prior work on any individually comparable metric; the contribution of this work is methodological – introducing a harder, theoretically motivated evaluation protocol to a setting where it is largely absent, and providing rigorous, honestly reported negative evidence about what does and does not help under that protocol.

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